

ECON 21030: Honors Econometrics

Practice Midterm 2

Spring 2023

There are a total of **120 points** on the exam.

1: *[35 points total]* You are interested in the effect of a binary treatment on an outcome of interest. To this end, you collect an iid sample of n individuals and assign them to the treatment or control group with equal probability. Let Y_i denote the outcome of the i th individual. It is assumed that $E[Y^2] < \infty$. Let D_i denote the treatment status of the i th individual, where $D_i = 1$ if the i th individual is treated and $D_i = 0$ if the i th individual is not treated. You assume the following model for Y_i :

$$Y_i = \alpha_i + \beta_i D_i$$

where (α_i, β_i) are independent of D_i . Note that α_i and β_i are random variables, and β_i is the effect of the treatment on the outcome. Furthermore the effect is "heterogeneous" in the sense that β_i differs across individuals.

1. *[5 points]* Explain why it might be reasonable to assume that (α_i, β_i) is independent of D_i .

Each individual is assigned with equal probability to the treatment and control group, therefore the individual characteristics that determine alpha and beta are independent of D_i .

2. *[10 points]* Consider the linear regression:

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

which you interpret as the best linear predictor of Y_i given D_i . Express α and β in terms of the distribution of (α_i, β_i) . In particular, show that $\beta = E[\beta_i]$

$$\beta = \frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\alpha_i + \beta_i D_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\alpha_i, D_i)}{\text{Var}(D_i)} + \frac{\text{Cov}(\beta_i D_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\beta_i D_i, D_i)}{\text{Var}(D_i)}$$

where the last equality follows (a)'s implication that $\text{Cov}(\alpha_i, D_i) = 0$.

$$\text{Cov}(\beta_i D_i, D_i) = E[\beta_i D_i^2] - E[\beta_i D_i] E[D_i] = E[\beta_i] (E[D_i^2] - E[D_i]^2) = E[\beta_i] (\text{Var}[D_i])$$

So $\beta = \frac{E[\beta_i](\text{Var}[D_i])}{\text{Var}(D_i)} = E[\beta_i]$. Then,

$$\alpha = E[Y_i] - \beta E[D_i] = E[\alpha_i + \beta_i D_i] - \beta E[D_i] = E[\alpha_i] + E[\beta_i D_i] - \beta E[D_i]$$

and since β_i is independent of D_i

$$= E[\alpha_i] + E[\beta_i] E[D_i] - \beta E[D_i] = E[\alpha_i] + \beta E[D_i] - \beta E[D_i] = E[\alpha_i]$$

3. [5 points] Do you think ε_i is heteroskedastic or homoskedastic? Why?

(hint: try writing $\varepsilon_i = \alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i$ and investigating $\text{Var} \varepsilon_i | D_i$ for both possible values of D_i .)

Following the hint:

$$\begin{aligned} \text{Var} \varepsilon_i | D_i &= \text{Var}[\alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i | D_i] \\ &= \text{Var}[\alpha_i + \beta_i D_i - E[\beta_i] D_i | D_i] \end{aligned}$$

which equals $\text{Var}[\alpha_i]$ if $D_i = 0$ and equals $\text{Var}[\alpha_i + \beta_i]$ if $D_i = 1$. Since these are not generally equivalent, the error term ε_i is most likely heteroskedastic.

4. [5 points] Define $X_i = (1, D_i)$ and define $\hat{\alpha}_n, \hat{\beta}_n$ as the OLS estimators of α and β given our sample. Recall from our usual asymptotic theory that we know:

$$\sqrt{n} \left(\begin{pmatrix} \hat{\alpha}_n, \hat{\beta}_n \end{pmatrix} - (\alpha, \beta)' \right) \xrightarrow{d} N(0, \Omega)$$

Write down an expression for Ω as well as a consistent estimator of Ω . Call this estimator $\hat{\Omega}_n$. You do not need to show that $\hat{\Omega}_n$ is consistent.

From the previous question, we know we must use heteroskedasticity robust standard errors. Therefore, we have:

$$\Omega = E[X_i X_i']^{-1} \text{Var}[X_i \varepsilon_i] E[X_i X_i']^{-1}$$

And:

$$\hat{\Omega}_n = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \varepsilon_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1}$$

5. [10 points] Suppose we are interested in obtaining a confidence interval around $\hat{\beta}_n$, but we do not care about $\hat{\alpha}_n$. Define $r' = (0, 1)'$ and show how to use this vector to derive a confidence interval for $\hat{\beta}_n$ at level $a \in (0, 1)$.

By the Continuous Mapping Theorem,

$$r' \hat{\Omega}_n r \xrightarrow{p} r' \Omega r$$

Also by the Continuous Mapping Theorem,

$$\begin{aligned} \sqrt{n} \left(r' \left(\hat{\alpha}_n, \hat{\beta}_n \right)' - r'(\alpha, \beta)' \right) &\xrightarrow{d} N(0, r' \Omega r) \\ \Rightarrow \sqrt{n} \left(\hat{\beta}_n - \beta \right) &\xrightarrow{d} N(0, r' \Omega r) \end{aligned}$$

Slutsky's and CMT gives us

$$\frac{\sqrt{n} \left(\hat{\beta}_n - \beta \right)}{\sqrt{r' \hat{\Omega}_n r}} \xrightarrow{d} N(0, 1)$$

Therefore we make our confidence region:

$$C_n := \left\{ c \in R : \frac{\sqrt{n} \left| \hat{\beta}_n - c \right|}{\sqrt{r' \hat{\Omega}_n r}} \leq z_{1-a/2} \right\}$$

where $z_{1-a/2}$ is the $1 - a/2$ quantile of standard normal distribution.

- 2:** [45 points total] Consider an i.i.d. sample $\{Y_i, D_i\}_{i=1}^n$ where Y_i is future earnings and $D_i \in \{0, 1\}$ indicates whether an individual received a scholarship to attend college. Let $Y_i(0)$ and $Y_i(1)$ represent the potential outcomes corresponding to the events $D_i = 0$ and $D_i = 1$, respectively.

1. [10 points] Suppose we write down the structural model:

$$Y_i = \beta_0 + \beta_{1,i} D_i + U_i$$

Where we assume $\mathbb{E}(U_i) = 0$ and define $\beta_0 = E[Y_i(0)]$. Provide an expression for the $\beta_{1,i}$ and U_i in terms of the potential outcomes and their first moments. How would you interpret the random variable $\beta_{1,i}$? Is it identified for any individual i ? Explain.

By definition, we have:

$$\begin{aligned} Y_i &= Y_i(0)(1 - D_i) + Y_i(1)D_i \\ &= Y_i(0) + \underbrace{(Y_i(1) - Y_i(0))}_{:=\beta_{i1}} D_i \end{aligned}$$

So, given our definition for $\beta_0 = E[Y_i(0)]$, U_i can be written as:

$$U_i = Y_i - \beta_0 - \beta_{i1}D_i$$

Which, again using the definition of Y_i simplifies to:

$$U_i = Y_i(0) - \mathbb{E}[Y_i(0)]$$

So we have the interpretation of U_i as capturing the individual variation in $Y_i(0)$.

β_{i1} captures the heterogeneous treatment effect of receiving the scholarship on future earnings. It is not identified for any individual i . This is because at the population level, each individual gets a single draw of $\{Y_i, D_i\}$, so we can never observe the counterfactual world in which the individual received a different assignment of D_i , even with all the population data. We can at best hope to recover the $\beta = \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]$.

2. [10 points] Is $\kappa = \mathbb{E}(Y_i|D_i = 1) - \mathbb{E}(Y_i|D_i = 0)$ identified? When is κ equal to ATT , $ATUT$, or ATE ?

κ is identified, since we observe Y_{ij} for all individuals assigned $D_i = j$.

Define:

$$\begin{aligned} ATT &\equiv \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ ATUT &\equiv \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 0] \\ ATE &\equiv \mathbb{E}[Y_i(1) - Y_i(0)]. \end{aligned}$$

Without further restrictions none of the above are identified.

If $(Y_i(1), Y_i(0)) \perp D_i, \forall i \in \{1, \dots, n\}$, then we can write:

$$\begin{aligned} \kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] \\ &= ATE \end{aligned}$$

Where we use the fact that $Y_i(0)$ and $Y_i(1)$ are independent of D_i to change the conditional expectations into unconditional expectations. Similarly, we can write:

$$\begin{aligned} \kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \mathbb{E}[Y_i(1)|D_i = 0] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= ATUT \end{aligned}$$

With an analogous expression clearly holding for ATT , so we see that $ATE = ATT = ATUT = \kappa$.

It is also easy to see that $ATE = ATT = ATUT = \kappa$ if $Y_i(1) - Y_i(0) = c$ a constant, since we have:

$$\mathbb{E}[c|D_i = 1] = \mathbb{E}[c] = \mathbb{E}[c|D_i = 0]$$

It is also possible that κ recovers only ATT or only $ATUT$. Specifically, if $\mathbb{E}[Y_i(0)|D_i = 0] = \mathbb{E}[Y_i(0)|D_i = 1]$ then we have that κ recovers ATT , and if $\mathbb{E}[Y_i(1)|D_i = 0] = \mathbb{E}[Y_i(1)|D_i = 1]$ then we have that κ recovers $ATUT$:

$$\begin{aligned}\kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] = \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1] = ATT \\ \kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] = \mathbb{E}[Y_i(1)|D_i = 0] - \mathbb{E}[Y_i(0)|D_i = 0] = ATUT\end{aligned}$$

Where, of course, if both statements hold we have the same result as if D_i were randomly assigned.

3. [10 points] Suppose the scholarship is only given to high-achieving students, who are already more likely to have higher earnings. Will κ overstate or understate the ATT and the $ATUT$?

The assumption we make here, from the problem statement, can be written as:

$$\mathbb{E}[Y_{ij}|D_i = 1] > \mathbb{E}[Y_{ij}|D_i = 0] \quad \text{for } j \in \{0, 1\}$$

In English: that high-achieving students (those for whom $D_i = 1$) have higher expected income than lesser-achieving students with the same scholarship status. This lets us write:

$$\begin{aligned}\kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \underbrace{\mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1]}_{ATT} + \underbrace{\mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]}_{>0 \text{ by assumption}}\end{aligned}$$

Meaning that κ overstates ATT . Similarly:

$$\begin{aligned}\kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \underbrace{\mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(1)|D_i = 0]}_{>0 \text{ by assumption}} + \underbrace{\mathbb{E}[Y_i(1)|D_i = 0] - \mathbb{E}[Y_i(0)|D_i = 0]}_{ATUT}\end{aligned}$$

Meaning that κ also overstates $ATUT$.

4. [10 points] Suppose the scholarship is given based on financial need, so that its recipients are more positively affected by D_i than other students are. How do you expect the ATT , $ATUT$, and ATE will compare in magnitude? Provide intuition behind your comparisons.

From the problem statement, we assume:

$$\mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1] > \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 0]$$

I.e., the students who are treated have a larger expected increase in their wages from receiving the scholarship. Since the LHS above is exactly the ATT and the RHS is exactly the ATUT, we clearly have that $ATT > ATUT$. Further notice that:

$$\begin{aligned} ATE &= \mathbb{E}[Y_i(1) - Y_i(0)] \\ &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1] \mathbb{E}[D_i]}_{ATT} + \underbrace{\mathbb{E}[Y_i(1) - Y_i(0)|D_i = 0] (1 - \mathbb{E}[D_i])}_{ATUT} \end{aligned}$$

So, assuming $\mathbb{E}[D_i] \in (0, 1)$, so that we have non-trivial treatment and control groups, we see that ATE is a weighted average of ATT and ATUT. So, we conclude:

$$ATT > ATE > ATUT$$

The intuition here is that if we selectively treat those who will benefit the most, the average treatment effect on them will be (mechanically) larger than on those we do not treat. As a result, the average treatment will fall somewhere between the treatments of the individual groups.

5. [5 points] Suppose that treatment D_i is randomized so that $\Pr(D_i = 1) = 0.5$. Show that $\kappa = ATE$.

Since the treatment is randomized we have that $(Y_i(1), Y_i(0)) \perp D_i$.

$$\begin{aligned} \kappa &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \mathbb{E}[Y_i(1) - Y_i(0)] \end{aligned}$$

which follows because randomized treatment implies that $\mathbb{E}[Y_i(1)|D_i = 1] = \mathbb{E}[Y_i(1)|D_i = 0]$ and $\mathbb{E}[Y_i(0)|D_i = 0] = \mathbb{E}[Y_i(0)|D_i = 1]$. Note that the probability does not matter. So long as $\Pr\{D_i = 1\} \in (0, 1)$ we have that OLS would recover the ATE.

- 3: [40 points total] Consider the following model of supply and demand:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D P + U^D & (\text{Demand}) \\ Q^S &= \beta_0^S + \beta_1^S P + U^S & (\text{Supply}) \end{aligned}$$

Throughout this exercise, assume that $E(U^D) = E(U^S) = E(U^D U^S) = 0$, and suppose you observe the market in equilibrium (i.e. $Q^D = Q^S$). Your goal is to estimate β_1^D and β_1^S .

1. [10 points] Show that P is endogenous in both equations.

Since the market is observed in equilibrium, setting $Q^D = Q^S$ gives:

$$\beta_0^D + \beta_1^D P + U^D = \beta_0^S + \beta_1^S P + U^S \Rightarrow P^* = \frac{(\beta_0^S - \beta_0^D) + (U^S - U^D)}{\beta_1^D - \beta_1^S}$$

Because the equilibrium price P^* depends on both U^S and U^D , we have that $E(P^*U^S) \neq 0$ and $E(P^*U^D) \neq 0$. Therefore, the price P^* is endogenous in both equations.

2. [10 points] Suppose you observe a variable Z that affects only the supply, so that:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D P + U^D \\ Q^S &= \beta_0^S + \beta_1^S P + \beta_2^S Z + U^S \end{aligned}$$

Assume that $E(ZU^D) = E(ZU^S) = 0$ and $\beta_2^S \neq 0$. Show that Z satisfies both instrument exogeneity and instrument relevance. Do these properties hold if Z also affects demand?

Note that Z satisfies instrument exogeneity because $E(ZU^D) = 0$ by assumption. To show that instrument relevance is satisfied, we write the following:

$$\text{Cov}(P, Z) = \text{Cov}\left(\frac{(\beta_0^S - \beta_0^D) + \beta_2^S Z + (U^S - U^D)}{\beta_1^D - \beta_1^S}, Z\right) = \frac{\beta_2^S}{\beta_1^D - \beta_1^S} \text{Var}(Z) \neq 0$$

If Z also affects demand, then we write: $Q^D = \beta_0^D + \beta_1^D P + \beta_2^D Z + U^D$, where $\beta_2^D \neq 0$. So:

$$Z = \frac{(\beta_0^S - \beta_0^D) + (\beta_1^S - \beta_1^D) P + (U^S - U^D)}{\beta_2^D - \beta_2^S}$$

Since $U^S - U^D$ appears in this expression, the instrument exogeneity assumption is no longer satisfied: $E(ZU^D) \neq 0$ and $E(ZU^S) \neq 0$. Of course, instrument relevance may hold.

3. [10 points] Using the setup from (b), express both P and Q as linear functions of Z . Then, assuming that Z is a valid instrument, show how to consistently estimate β_1^D from data $\{Q_i, P_i, Z_i\}_{i=1}^n$.

After re-arranging the supply-demand equations in part (b), we can write:

$$\begin{aligned} P &= \frac{\beta_0^S - \beta_0^D}{\beta_1^D - \beta_1^S} + \frac{\beta_2^S}{\beta_1^D - \beta_1^S} Z + \frac{U^S - U^D}{\beta_1^D - \beta_1^S} \\ Q &= \frac{\beta_1^D \beta_0^S - \beta_0^D \beta_1^S}{\beta_1^D - \beta_1^S} + \frac{\beta_1^D \beta_2^S}{\beta_1^D - \beta_1^S} Z + \frac{U^S \beta_1^D - U^D \beta_1^S}{\beta_1^D - \beta_1^S} \end{aligned}$$

Also, given the setup from (b), we can write down the IV estimator:

$$\hat{\beta}_1^{IV} = \frac{\sum_{i=1}^n (Z_i - \bar{Z}_n) (Q_i - \bar{Q}_n)}{\sum_{i=1}^n (Z_i - \bar{Z}_n) (P_i - \bar{P}_n)} = \frac{\sum_{i=1}^n (Z_i - \bar{Z}_n) Q_i}{\sum_{i=1}^n (Z_i - \bar{Z}_n) P_i} \xrightarrow{p} \frac{\text{Cov}(Z, Q)}{\text{Cov}(Z, P)} = \beta_1^D$$

4. [10 points] Now, suppose that consumers in the market face a tax τ , so that:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D(P + \tau) + U^D \\ Q^S &= \beta_0^S + \beta_1^S P + U^S \end{aligned}$$

Assume $\tau \perp (U^D, U^S)$. Given $\{Q_i, P_i, \tau_i\}$, show how to consistently estimate β_1^D and β_1^S . How would you interpret the coefficients β_1^D and β_1^S ? Justify your response.

The price consumers actually pay is $P^c = P + \tau$. Write the first IV estimator as:

$$\hat{\beta}_{\tau}^{IV,D} = \frac{\sum_{i=1}^n (\tau_i - \bar{\tau}_n) Q_i}{\sum_{i=1}^n (\tau_i - \bar{\tau}_n) P_i^c} \xrightarrow{p} \frac{\text{Cov}(Q, \tau)}{\text{Cov}(P^c, \tau)} = \beta_1^D$$

Write the second IV estimator to be:

$$\hat{\beta}_{\tau}^{IV,S} = \frac{\sum_{i=1}^n (\tau_i - \bar{\tau}_n) Q_i}{\sum_{i=1}^n (\tau_i - \bar{\tau}_n) P_i} \xrightarrow{p} \frac{\text{Cov}(Q, \tau)}{\text{Cov}(P, \tau)} = \beta_1^S$$

We can interpret $-\beta_1^D$ as the elasticity of demand and β_1^S as the elasticity of supply.